URL Safety Checker

Project description URL Safety Checker is a beginner-friendly cybersecurity mini-project. It examines a URL's format and highlights common warning signs that may appear in suspicious links.

Features - Checks whether the URL format appears valid. - Checks whether HTTPS is used. - Displays the detected hostname. - Flags a few warning signs, such as an unusually long URL, many hyphens, an IP-address hostname, an `@` symbol in the address, and words sometimes used in misleading links. - Prints a simple report in the terminal.

Technology used - Python 3 - Python standard library: `urllib.parse` and `re`

Dataset This project does not require an external dataset. You can test it with URLs you already know, or with harmless example URLs. Do not open suspicious links.

Setup and run 1. Install Python 3. 2. Save the program as `SathishKumar_URLChecker.py`. 3. Open a terminal or command prompt in the folder containing the file. 4. Run:

`python SathishKumar_URLChecker.py`

5. Enter a URL when prompted.

Example test Input: `https://example.com`

The program should report that the format appears valid and that HTTPS is used. The exact warnings depend on the URL entered.

Limitations and safety This project uses simple rules only. It does not check a website's reputation, scan files, detect malware, or guarantee that a website is safe. A URL with no warnings can still be malicious. Never enter passwords or personal information on an unfamiliar website.

Files - `SathishKumar_URLChecker.py` — project source code - `requirements.txt` — dependency list - `SathishKumar_ProjectReport.docx` — project report - `README.md` — project overview and run instructions